Assignment-02 (Lab-2.2)

Course : AI Assisted Coding

Topic:- Exploring Additional AI Coding Tools beyond Copilot – Gemini (Colab)and Cursor AI

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Task-01

Final Optimal Prompt

#write the python function to only negative number from use input list,output have before and after finding number

Code Screenshot

Output Screenshot

```
Original List: [1, -2, 3, -4, 0, -5, 6]
List with Negative Numbers: [-2, -4, -5]
0
```

Explanation/Justification/Observation (100 words / 5 – 6 sentence)

This function uses a list comprehension to create a new list that includes only the no negative values from the original list. It does not modify the original list,ensuring that the raw sensor data remains intact for any further analysis or processing.

Task-02

Final Optimal Prompt

#write the python function to print the count of vowels, consonants, digits in given user input string

Code Screenshot

```python
#write the python function to print the count of vowels, consonants, digits in given user input string
def count_characters(input_string):
    vowels = "aeiouAEIOU"
    vowel_count = 0
    consonant_count = 0
    digit_count = 0

    for char in input_string:
        if char.isalpha():  # Check if the character is an alphabet
            if char in vowels:
                vowel_count += 1
            else:
                consonant_count += 1
        elif char.isdigit():  # Check if the character is a digit
            digit_count += 1

    print("String:", input_string)
    print("Vowels:", vowel_count)
    print("Consonants:", consonant_count)
    print("Digits:", digit_count)

# Example usage:
my_string = "Hello World 123"
count_characters(my_string)

another_string = "Python is fun! 4ever"
count_characters(another_string)
```

Output Screenshot

```
String: Hello World 123
Vowels: 3
Consonants: 7
Digits: 3
String: Python is fun! 4ever
Vowels: 5
Consonants: 10
Digits: 1
```

Explanation/Justification/Observation (100 words / 5 – 6 sentence)

This function iterates through each character in the input string, checking if it is a vowel, consonant, or digit, and increments the respective counters accordingly. It treats upper- and lower-case letters the same by including both in the vowels string and using is alpha() for consonants.

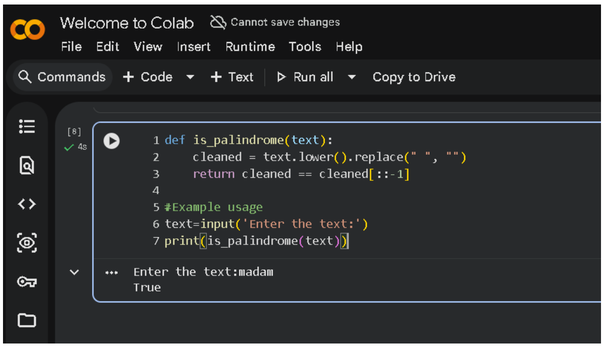
Task-03

Final Optimal Prompt

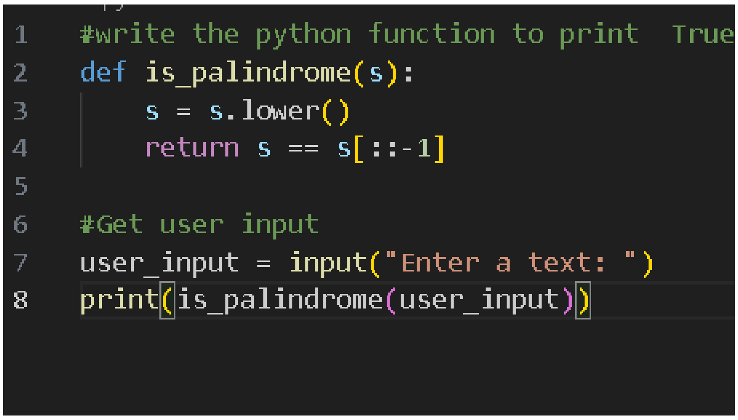
#write the python function to print True or False the given user input whether given text is "palindrome" or not

Code Screenshot

Copilot screen shot:



Gemini Screen shot:



Output ScreenshotExplanation/Justification/Observation (100 words / 5 – 6 sentence)

Criteria	Gemini	GitHub Copilot
Logic Clarity	Uses intermediate variable (cleaned), making each step clear and easy to understand.	Uses compact logic with fewer lines, which is concise but slightly less descriptive.
Readability	High readability due to meaningful variable naming and explicit preprocessing.	Readable for experienced programmers but may be less clear for beginners.

Handling Input Variations	Removes spaces and normalizes case, making it more robust for real-world strings.	Handles only case conversion; does not remove spaces or special characters.
Beginner Friendliness	More beginner-friendly due to step-by-step logic.	Better suited for users already comfortable with Python slicing.
Code Conciseness	Slightly longer but clearer.	Very concise and minimal.
Maintainability	Easy to modify (e.g., add punctuation removal).	Requires more understanding to extend functionality.

Task-04

Final Optimal Prompt

Python function either a prime checker or palindrome checker and explain it line by line in simple, student friendly language, as if teaching a beginner.

Code Screenshot

```python
#Task4
#Python function either a prime checker or palindrome checker and e
def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n ** 0.5) + 1):
        if n % i == 0:
            return False
    return True
#Example usage:
number_to_check = 29
is_number_prime = is_prime(number_to_check)
print(f'Is the number {number_to_check} prime? {is_number_prime}')
```

Output Screenshot

```
Is the number 29 prime? True
```

Explanation/Justification/Observation (100 words / 5 – 6 sentence)

The function checks if a number is prime by first handling numbers less than 2 (which are not prime). Then it iterates from 2 up to the square root of the number, checking for divisibility. If any divisor is found, it returns False; otherwise, it returns True.